

Unit 116 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 89.2 ac, BAF 20 variable-radius cruise on a 300 ft grid, 44 plots, 507 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	230 sq ft/ac (SE 1.5 %)	Type	Sierran mixed conifer
Trees per acre	70	CWHR size / density	5 / D (cover 94 %)
QMD	24.6 in	SDI	283 (49 % of max 573)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	230 sq ft/ac 24.9 in	BA QMD	1 sq ft/ac 9.4 in
TPA	68	TPA	2
Volume	53.6 MBF/ac (9740 cu ft)	Volume	0.7 green tons/ac (22 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~163 sq ft/ac BA	Sampling error	3.1 % at 95 % (t = 2.02, n = 44)
Species (BA)	PP 65, WF 62, DF 42, IC 36, SP 25	Stratum standard	40 % - MEETS; 44 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	6	4.9	2.7	64	0.4
12-15.9	27	10.5	12.3	362	2.0
16-19.9	57	14.7	25.9	847	4.7
20-23.9	71	12.3	32.3	1154	6.3
24-27.9	86	10.7	39.1	1571	8.6
28-31.9	70	6.4	31.8	1407	7.7
32-35.9	58	4.3	26.4	1212	6.7
36-39.9	44	2.6	20.0	985	5.4
40-43.9	29	1.4	13.2	686	3.8
44-47.9	15	0.6	6.8	352	1.9
48-51.9	21	0.7	9.5	520	2.9
52-55.9	10	0.3	4.5	253	1.4
56-59.9	5	0.1	2.3	132	0.7
60-63.9	8	0.2	3.6	218	1.2
All	507	69.7	230.5	9763	53.7

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	144	17.6	65.5	26.1	2802	15.4
WF	136	19.6	61.8	24.0	2622	14.4
DF	93	13.1	42.3	24.4	1766	9.7
IC	79	11.8	35.9	23.6	1482	8.2
SP	55	7.6	25.0	24.6	1089	6.0
All	507	69.7	230.5	24.6	9763	53.7

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
116-01	10	200	17	ok
116-02	12	240	17	ok
116-03	11	220	11	ok
116-04	10	200	20	ok
116-05	11	220	24	ok
116-06	12	240	15	ok
116-07	10	200	14	ok
116-08	10	200	11	ok
116-09	12	240	14	ok
116-10	11	220	19	ok
116-11	11	220	21	ok
116-12	14	280	12	ok
116-13	11	220	10	ok
116-14	11	220	4	ok
116-15	12	240	3	ok
116-16	10	200	15	ok
116-17	13	260	12	ok
116-18	11	220	16	ok
116-19	12	240	20	ok
116-20	12	240	12	ok
116-21	11	220	8	ok
116-22	10	200	4	ok
116-23	11	220	12	ok
116-24	13	260	12	ok
116-25	11	220	20	ok
116-26	11	220	14	ok
116-27	13	260	11	ok
116-28	13	260	16	ok
116-29	13	260	27	ok
116-30	11	220	22	ok
116-31	12	240	20	ok
116-32	14	280	12	ok
116-33	11	220	6	ok
116-34	10	200	10	ok
116-35	11	220	14	ok
116-36	11	220	11	ok
116-37	12	240	11	ok
116-38	13	260	13	ok
116-39	12	240	9	ok
116-40	13	260	10	ok
116-41	9	180	5	ok
116-42	11	220	11	ok
116-43	13	260	4	ok
116-44	12	240	19	ok